

Adding Swap Partitions

Using swap on Linux is a convenient way to improve Linux kernel memory usage. If a shortage of physical RAM occurs, non-recently-used memory pages can be moved to swap, which makes more RAM available for programs that need access to memory pages.

1. Use `fdisk /dev/sdb` to open your disk in `fdisk`
2. Press `n` to add a new partition. Specify start and stop cylinders and size.
3. Type `t` to change the partition type. If you are using `fdisk`, use partition type 82.
If you are using `gdisk`, use partition type 8200.
4. Use `mkswap` to format the partition as swap space. Use, for instance, `mkswap /dev/sdb1` if the partition you have just created is `/dev/sdb1`.
5. Type `free -m`. You see the amount of swap space that is currently allocated.
6. Use `swapon` to switch on the newly allocated swap space. If, for instance, the swap device you have just created is `/dev/vda6`, use `swapon /dev/vda6` to activate the swap space.
7. Type `free -m` again. You see that the new swap space has been added to your server.

Mounting File Systems

Mounting process is to make file system accessible through creating mount point

In windows mounting process is under letters like (c: - d: - e:etc) but in linux you can mount any partition or file system under directory (folder) so you have to create mount point (`mkdir /mount`).

Types of mounting :

- Temporary (in running time and still take effect as long as system up and running but when you reboot the system everthing is gone.
- Permanent through adding entries in `/etc/fstab` file.

To mount a file system, some information is needed:

■ **What to mount:** This information is mandatory and specifies the name of the device that needs to be mounted.

■ **Where to mount it:** This is also mandatory information which specifies the directory on which the device should be mounted.

■ **What file system to mount:** Optionally, you can specify the file system type. In most cases, this is not necessary. The **mount** command will detect which file system is used on the device and make sure the correct driver is used.

■ **Mount options:** Many mount options can be used when mounting a device. Using options is optional and depends on the needs you may have with the file system.

Manually Mounting File Systems

Mounting >>> `mount /dev/vda5 /mount`

unmounting >>> `umount /dev/vda5` or `umount /mount`

You can mount the partition through :

1. Device name
2. Label
3. UUID (best way to mount)

To make Label to a partition :

For ext4 file system

```
#e2label /dev/sdb1 data
```

```
#mount LABEL=data /mount
```

For xfs file system

```
#xfs_admin -L sales /dev/sdb1
```

```
#mount LABEL=sales /dev/sdb1
```

Mounting through UUID:

After you apply file system to a partition the system generate a UUID for it and it's unique

You can use the **blkid** command to get an overview of the current file systems on your system and the UUID that is used by that file system.

```
[root@server3 ~]# blkid
/dev/vda1: UUID="02305166-840d-4f74-a868-c549b3229e65" TYPE="xfs"
/dev/vda2: UUID="Hasz5q-nZF3-XN94-L2fm-xUwz-3VEq-qVCAYi" TYPE="LVM2_
member"
/dev/vda5: UUID="42f419c4-633f-4ed7-b161-519a4dadd3da" TYPE="xfs"
/dev/mapper/centos-swap: UUID="5867ba02-fd89-475c-be56-7922febde43b"
TYPE="swap"
/dev/mapper/centos-root: UUID="b2022ac4-73c6-4e6b-a52f-703e3e2476b7"
TYPE="xfs"
```

```
mount UUID="42f419c4-633f-4ed7-b161-519a4dadd3da" /mount
```

Permanent Mount Through /etc/fstab

In the /etc/fstab file, everything is specified to mount the file system automatically. For this purpose, every line has six fields.

Field	Description
Device	The device that must be mounted. A device name, UUID, or label can be used.
Mount Point	The directory or kernel interface where the device needs to be mounted.
File System	The file system type.
Mount Options	Mount options.
Dump Support	Use 1 to enable support to backup using the dump utility. This may be necessary for some backup solutions.
Automatic Check	Specifies if the file system should be checked automatically when booting. Use 0 to disable automated check, 1 if this is the root file system and it has to be checked automatically, and 2 for all other file systems that need automatic checking while booting. Network file systems should have this option set to 0.

Mount Options

Option	Use
auto/ noauto	The file system will [not] be mounted automatically.
acl	Adds support for file system access control lists (see Chapter 7, “Configuring Permissions”).
user_xattr	Add support for user extended attributes (see Chapter 7).
ro	mounts the file system in read-only mode.
atime / noatime	Disables or enables access time modifications.
noexec / exec	Denies or allows execution of program files from the file system.
_netdev	Use this to mount a network file system. This tells fstab to wait until the network is available before mounting this file system.

Open /etc/fstab in an editor and add the following line:

```
UUID="nnnn" /mounts/data xfs defaults 1 2
```

```
#mount -a
```

Eng. Muhammad Adel